Insights into the biological bases of modern human brain shape

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Modern humans have a globular brain shape, a feature which is unique in the hominin fossil record and estimated to have emerged within the *Homo sapiens* lineage between 100,000 and 35,000 years ago (Neubauer et al., 2018). This timing coincides with the appearance of key components of modern human behaviour (e.g., complex technologies, expanded social networks) and the global dispersal of *H. sapiens*. Although these associations are correlational, they motivate targeted investigation of the biological mechanisms underlying endocranial globularity. Moreover, identifying recently evolved developmental pathways that produce human-specific traits may also shed light on vulnerability to, and mechanisms of, human brain disorders.

Insights into genetic foundations can be gained by neuroimaging genomics using variation in present-day humans in sufficiently large sample sizes. Using geometric morphometric methods, we quantified endocranial shape based on 935 landmarks and semi-landmarks on MRI brain scans from the UK Biobank (Sudlow et al., 2015; Miller et al., 2016). From these data we computed a scalar metric for globularity, where high values represent more globular endocrania and low values more elongated endocrania, with minimal overlaps between modern human variation and Neandertal/*Homo erectus* variation (see Fig. 1). A genome wide association study (GWAS) on this globularity score in a sample of 33,000

individuals found associations with 28 loci and 77 genes, and a shared genetic basis with the ventricular system, the superior temporal sulcus and white matter microstructure (Molz et al., 2024). The genetic variants underlying globularity were also enriched for genes expressed in the brain during the perinatal period and early childhood. Notably, Molz et al. (2024) identified a significant genetic correlation with a reading/language-related phenotype (Eising et al., 2022), while genetic correlations with cognitive skills (Demange et al., 2021) were not significant. We investigated the most significant locus of the globularity GWAS, associated with the gene *C16orf95*, and found that it is close to a differentially methylated region in archaic hominins, is implicated in the connection between the superior temporal and the middle temporal lobe (Sha et al., 2023), and when deleted leads to the rare condition macrocerebellum, which leads to parietal bulging (Seeley et al., 2014).

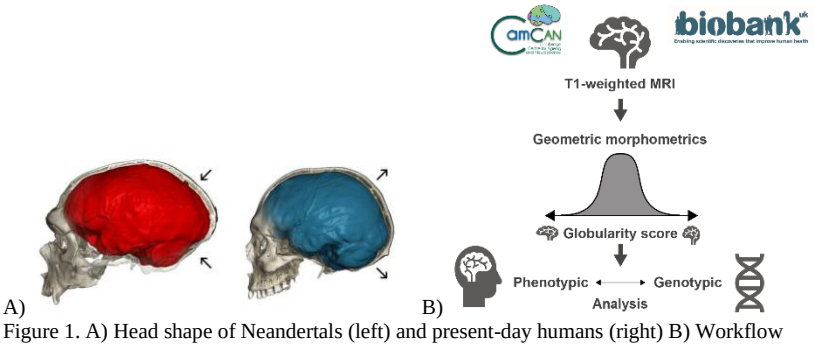

Figure 1. A) Head shape of Neandertals (left) and present-day humans (right) B) Workflow

Our current focus is on rare variants in genes due to their potential higher effect size and better biological interpretability, as compared to common variants. We aggregate the effects of protein-altering rare variants (MAF<0.001) on each gene in ~80,000 individuals from the UK Biobank, and test for their associations with endocranial globularity using burden analysis implemented in Regenie (Mbatchou et al., 2021). We assess genes showing rare-variant associations with endocranial globularity for overlap with prior GWAS results (Molz et al., 2024) and gene sets related to cortical layers, white matter structures, and brain development. Furthermore, we assess overlap with genes co-located with Neandertal introgressed fragments, genes in Neandertal introgression deserts (Buisan et al., 2022), as well as genes co-located with common variants of ventricular morphology (Ge et al., 2024). By elucidating the biology underlying endocranial globularity, we hope to better understand this recent and uniquely modern human trait and its potential connections to language.

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